

Evidence-Constrained Diagnostic Assistance for Unlabeled Neutron-Current Time Series: A Tool-Using Workflow and Retrieval Robustness Study

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Abstract

Neutron-instrumentation records from nuclear power plants are high-volume raw time series for which exhaustive, context-sensitive expert fault labeling is impractical at archive scale, whereas their interpretation must remain traceable to authoritative technical evidence. We present an evidence-constrained diagnostic-assistance workflow rather than an autonomous fault classifier. A zero-prior, tool-using agent first converts unlabeled neutron-current CSV records into an auditable descriptive condition report by applying deterministic probes for data health, morphology, anomalies, lag, extremes, and spatial snapshots. A compact structured alert summary then guides human-specified, multi-stage retrieval over nuclear instrumentation and control manuals, enabling a citation-bearing candidate-interpretation memo. We evaluate the retrieval layer under controlled expansion of a fixed IAEA core library with same-domain supplementary documents, using Chinese and English queries, chunking configurations, and both strict chunk and document-level relevance criteria. In the evaluated setup, cosine similarity can increase while the priority of pre-designated IAEA sources decreases; the primary query-formulation-by-chunking comparison (P0) further shows that query family, language, encoder, and chunking interact rather than yielding a universal query ranking. The results show that similarity alone is insufficient to govern evidence retrieval in this setting and motivate explicit source-priority policies, technically specific query formulation, and human review of diagnostic conclusions.

Keywords: neutron instrumentation, online monitoring, diagnostic assistance, retrieval-augmented generation, evidence traceability, retrieval robustness

1. Introduction

Nuclear power plants continuously acquire neutron and instrument-channel measurements for surveillance and condition monitoring. These raw time series can reveal changes in detector behavior, signal integrity, and operating regimes, but exhaustively assigning reliable fault labels to long, multichannel archives requires context-sensitive expert review with operating and maintenance information. That effort is impractical at routine archive scale, and the CSV records used here consequently do not provide a uniform sample-level fault ground truth. Technical interpretation therefore continues to rely on authoritative online-monitoring and instrument-channel-assessment guidance^{1–3}. Neutron-noise techniques provide a non-intrusive route to studying reactor and instrumentation phenomena⁴, while projects such as CORTEX extend numerical simulation and validation capabilities for reactor monitoring⁵. Data-driven studies have demonstrated scenario-specific neutron-detector validation and nuclear-plant fault classification^{6,7}; however, the fixed signal constructions and predefined fault settings used in such studies are not always available in unlabeled plant archives.

Large language models (LLMs) can help organize heterogeneous observations and technical knowledge, but free-form generation is unsuitable as a decision authority in a safety-relevant setting. Retrieval-augmented generation (RAG)⁸ conditions an answer on external evidence; in nuclear applications, its value depends on whether the retrieved manuals genuinely support the statement being made. Nuclear-domain studies use retrieval and factuality checks to improve the reliability of technical explanations^{9,10}. Document-interpretation experiments identify failure modes including incorrect retrieval, unsupported interpretation, hallucination, and refusal¹¹, while RADIANT-LLM emphasizes provenance tracking, citation-backed responses, and human validation¹². A practical open question is how to connect measurement-side findings to that evidence layer without asking an LLM to infer a root cause directly from a raw CSV file.

This work addresses that question through an evidence-constrained diagnostic-assistance workflow. A tool-using agent observes unlabeled raw neutron-current series with minimal diagnostic priors and writes an auditable descriptive condition report. This report is a derived analysis artifact, not a ground-truth annotation: it records signal phenomena such as spikes, zero events, co-movement, lag, and data-health cues without promoting them to a fault

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class. It is then compressed into a structured alert summary that guides human-specified retrieval and citation-bearing explanation. The intended output is not an automatically confirmed fault diagnosis: it separates observed data phenomena, manual-supported candidate interpretations, and the engineering review that remains necessary to judge either.

The reliability of this workflow depends critically on retrieval. A manual library may need to include both a small, pre-designated authoritative set and a larger set of same-domain supplements. Adding relevant supplements can improve apparent nearest-neighbor similarity while reducing the priority of the authoritative sources specified by a deployment policy. This tension is especially important for bilingual technical terminology and for overlap-based chunking, where a useful passage may not have the identical chunk identifier used as a query probe.

The contributions are threefold. First, we develop a zero-prior, tool-using workflow with a persistent workspace contract for converting neutron-current CSV data into auditable descriptive observations, providing a scalable intermediate layer when exhaustive fault labeling is unavailable. Second, we introduce report-driven multi-stage RAG that maps structured observations to citable manual evidence. Third, we evaluate evidence retrieval under controlled core-supplement expansion, query language, chunking, and strict versus document-level relevance. The resulting evidence is used to characterize deployment boundaries rather than to claim autonomous fault identification. The complete transformation contract is specified in section 2.

2. System Design and Methodology

This section specifies the evidence-constrained transformation contract used throughout the study: a raw measurement record is converted into an auditable descriptive observation packet, compressed into a structured alert summary, linked to retrieved manual evidence, and expressed as a citation-bearing memo for engineering review. The intermediate description makes large, unlabeled archives reviewable without treating the automated chain as a fault classifier; engineering review remains outside that chain and retains decision authority.

The complete transformation contract is summarized in fig. 1: it distinguishes the auditable measurement-side artifacts from retrieved documentary evidence and from the external engineering-review decision point.

2.1. Data Resources

The system consumes two complementary data resources: a plant neutron-current measurement archive that supplies observable signal phenomena, and a technical evidence corpus that supplies retrievable documentary support. Their roles are deliberately distinct: neither resource alone is treated as a fault-label database.

Evidence-constrained diagnostic assistance

A report-centered architecture that separates observed signal events, documentary evidence, and engineering judgment.

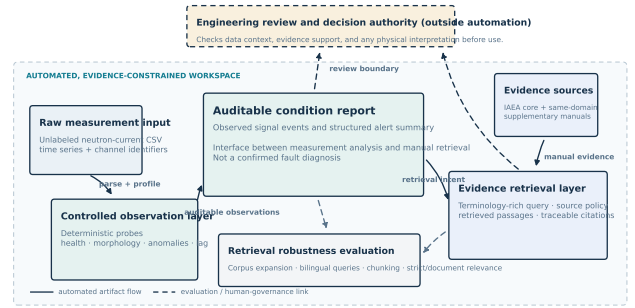


Figure 1: Evidence-constrained diagnostic-assistance architecture.

2.1.1. Neutron-current measurement archive

The measurement archive contains 42 multi-channel neutron-current CSV records from four instrumentation-bank groups (A1, A2, B1, and B2) in the same plant campaign. Each record contains one time column and seven neutron-current channels (eight columns overall). The archive composition is summarized in table 1.

Table 1: Composition of the neutron-current measurement archive.

Group	Number of series	Channels per series	Columns per series
A1	11	7	8
A2	11	7	8
B1	10	7	8
B2	10	7	8
Total	42	—	—

Channel identifiers follow plant tagging conventions (e.g., RIC...MA_VER). Within a group, the seven detectors occupy a circumferential layout, so their ordering in the CSV is associated with relative channel position rather than an arbitrary list. Representative series contain on the order of 6×10^5 samples at a nominal 1 s interval over multi-day windows.

The raw exports can contain local timestamp disorders and non-monotonic segments. Chronological sorting is therefore a mandatory first operation. More importantly, assigning a reliable fault label to every local pattern would require an expert to reconcile the time series with operating context, maintenance history, and instrumentation knowledge across the full archive. The present workflow therefore does not assume a uniform sample-level fault ground truth. Its output is a condition report of observed signal events and relationships, not a supervised fault class; derived reports are workflow artifacts rather than annotations attached to the raw CSV records.

2.1.2. Technical evidence corpus

The evidence resource is a curated 64-document collection of nuclear instrumentation and control manuals converted from PDF to Markdown and accompanied by

document-source metadata. It contains a 13-document IAEA core, including IAEA online-monitoring and instrument-channel-assessment guidance^{1,2}, together with 51 same-domain supplementary documents from sources such as NRC, EPRI/DOE, OECD/NEA, and laboratory or topical reports. The historical and independent embedding-model protocols use distinct index and chunk constructions, as detailed in section 3, but draw from this same document collection.

The supplementary documents are not assumed to be irrelevant. They enlarge the searchable knowledge base, while the IAEA core represents a deployment policy that prioritizes a pre-designated authority set. This distinction motivates the source-priority measurements in the evaluation but does not redefine source priority as universal relevance.

2.1.3. Data roles and semantic boundary

The two resources meet only through an auditable semantic interface. Deterministic probes first convert a measurement record $\mathbf{X}$ into observed events and a structured descriptive alert summary $\mathbf{a} = g(\mathbf{X})$. Human-specified query formulations then use those cue families to search the evidence corpus. Thus a measurement anomaly is not directly promoted to a physical diagnosis, and a manual passage is not treated as a direct explanation of an unprocessed waveform. The resulting evidence-linked memo remains an aid to engineering review.

2.2. Measurement-side observation construction

2.2.1. Objective and zero-prior scope

The measurement-side objective is to convert one raw neutron-current CSV record into a structured, human-readable descriptive condition report. “Zero-prior” does not mean that the system ignores the data format or the relative channel layout described in section 2.1. Rather, it does not assume a labeled fault library, a pre-specified mapping from an observed pattern to a physical cause, or handcrafted diagnostic frequency bands. This intermediate descriptive layer is deliberate: it allows large archives to be screened and connected to technical evidence without requiring every signal pattern to be manually fault-labeled in advance. The workflow treats channel identifiers as operational identifiers and uses deterministic statistics to expose morphology, redundancy, anomalies, lag, extremes, and cross-channel snapshots. Its output is an observation record, not a confirmed fault class.

2.2.2. Workspace initialization and data health

The load-and-profile probe is the mandatory entry point. Given a CSV path, it creates a dedicated workspace, parses and chronologically sorts the time column, writes a cleaned active record, and records shape, nominal sampling interval, missingness, and inactive-channel indicators in a persistent blackboard (`state.json`). All later probes resolve the active record from this workspace rather than accepting a

new free-text data path. This establishes one analysis unit per record and preserves the provenance of every subsequent observation.

The measurement-side transformation from one active record to an observation packet is summarized in fig. 2. Its microvisuals are representative outputs of registered deterministic probes; they illustrate observable signal evidence rather than confirmed fault labels.

2.2.3. Deterministic observation probes

After initialization, the planner may select only the registered probes in table 2. The discovery probe performs a global longitudinal scan: uniqueness and coefficient of variation assign coarse morphological roles, while pairwise correlations identify near-redundant pairs and strongly co-moving groups. Given known relative channel placement, such co-movement can motivate inspection of spatial proximity or shared instrumentation context; it is not, by itself, a physical diagnosis.

The anomaly/regime probe segments channels into steady periods and periods of short-duration change, and enumerates spike episodes and isolated zero events. The time-lag probe distinguishes lead-lag patterns from near-synchronous groups. Extremes are localized with their timestamps, and the snapshot probe records a multi-channel cross-section at a selected time. Together, these operations form a reproducible inventory of signal events and relationships.

Table 2: Deterministic probe families used to construct the condition report and their auditable outputs.

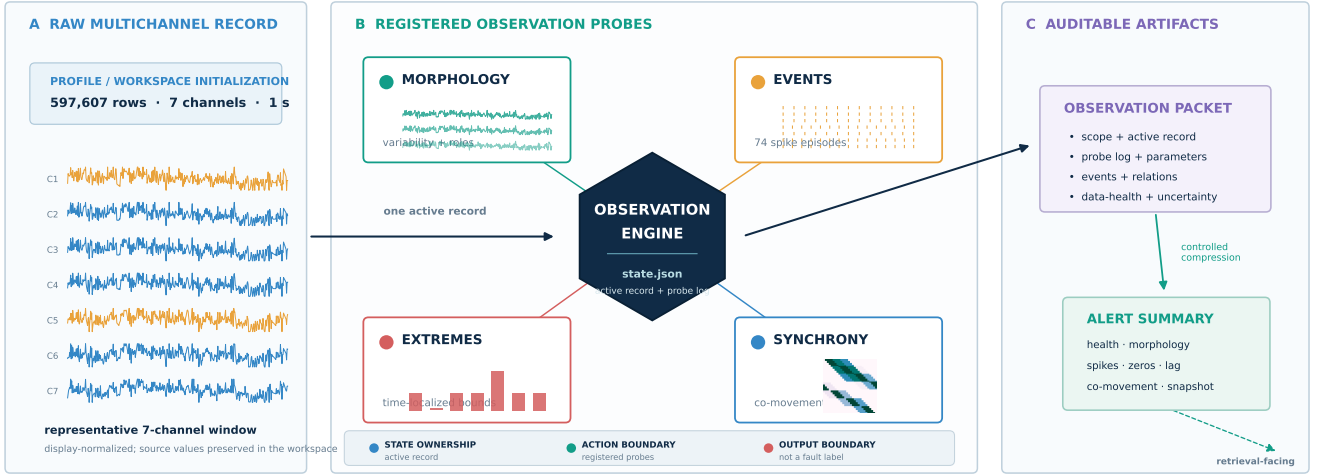
Probe family	Deterministic operation	Auditable output
Workspace initialization	Sorts timestamps, establishes the active record, and checks sampling, missingness, and inactive channels.	Active record and data-health indicators
Morphology and redundancy	Computes variability and pairwise relations to identify coarse roles and strongly co-moving or near-redundant groups.	Morphology roles and channel-relation groups
Event and regime scan	Segments steady periods and periods of short-duration change, and enumerates spike episodes and isolated zero events.	Regime segments and event inventory
Synchrony and lag scan	Estimates lead-lag structure and distinguishes it from near-synchronous co-movement.	Lag estimates and synchrony groups
Extremes	Localizes channel-wise maxima and minima in time.	Extreme values and timestamps
Cross-channel snapshot	Captures the multichannel state at a selected time.	Time-localized state vector

2.2.4. Operational definitions and auditable interface

Let $\mathbf{X} = \{x_{t,c}\}$ denote a chronologically sorted multi-channel record, where t indexes the sample and c indexes the detector channel. The system does not ask the planner

From multichannel record to auditable observation packet

Four registered probes turn one active record into traceable signal evidence before retrieval or interpretation.



Microvisuals are derived from a representative archived record and its workflow artifacts; they illustrate probe outputs, not confirmed fault labels.

Figure 2: Data-grounded measurement-side observation method: active record, registered probes, observation packet, and alert summary.

to estimate these quantities. Deterministic probes compute them and write their results to the workspace blackboard; the LLM only selects registered probes and organizes the resulting evidence.

For each channel, the anomaly probe uses a centered rolling window W_t of 31 samples. The local baseline and robust local deviation used by the spike rule in eq. (2) are defined in eq. (1).

$$\begin{aligned} m_{t,c} &= \text{median}_{\tau \in W_t} x_{\tau,c}, \\ d_{t,c} &= \text{median}_{\tau \in W_t} |x_{\tau,c} - m_{\tau,c}|. \end{aligned} \quad (1)$$

Because $d_{t,c}$ is recomputed at every t from the deviations within W_t , it adapts to local scatter rather than to the absolute current level: a more variable window produces a larger $d_{t,c}$ and hence a wider local threshold band. The two-condition spike decision is then specified in eq. (2):

$$\begin{aligned} I_{t,c}^{\text{spike}} &= \mathbb{I}(|x_{t,c} - m_{t,c}| > 5d_{t,c}) \mathbb{I}(|x_{t,c} - m_{t,c}| > 0.05R_c), \\ R_c &= \max_t x_{t,c} - \min_t x_{t,c}. \end{aligned} \quad (2)$$

Contiguous positive samples are merged into one spike episode. An isolated zero event is separately recorded by $I_{t,c}^{\text{zero}} = \mathbb{I}(x_{t,c} = 0)$. These rules define detected signal events, not physical fault labels. Figure 3 applies eqs. (1) and (2) to a representative 5,000-sample raw-record excerpt. Its shaded band changes with local scatter, and the orange line and emphasized red marker identify the largest rule-positive departure in that excerpt.

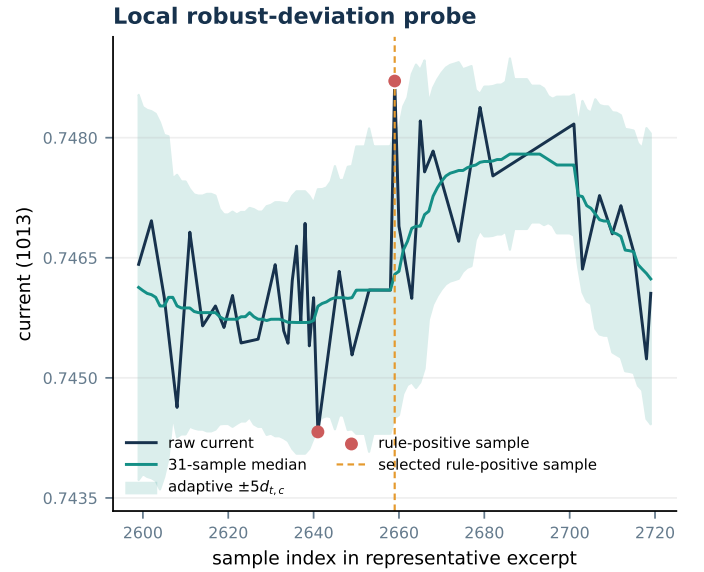


Figure 3: Representative robust-deviation calculation based on eqs. (1) and (2).

For a channel pair (i, j) , the discovery probe computes the Pearson coefficient defined in eq. (3).

$$r_{ij} = \frac{\sum_t (x_{t,i} - \bar{x}_i)(x_{t,j} - \bar{x}_j)}{\sqrt{\sum_t (x_{t,i} - \bar{x}_i)^2} \sqrt{\sum_t (x_{t,j} - \bar{x}_j)^2}}. \quad (3)$$

The current implementation flags near-redundancy at $r_{ij} \geq 0.99$ and strong co-movement for $0.85 \leq r_{ij} < 0.99$. Figure 4 visualizes eq. (3) for one selected pair in the same representative excerpt. It is a statistical co-movement

display, not evidence of an instrument relationship or physical mechanism. Timing is inspected separately by the correlation-lag scan in fig. 5; its peak is an association, not a causal direction. Together with lag, extremes, and spatial-snapshot outputs, these quantities form the structured alert interface in eq. (4).

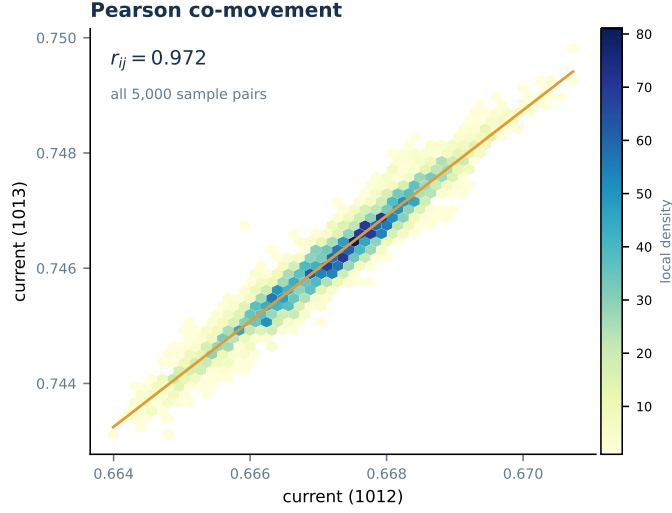


Figure 4: Representative Pearson co-movement density view based on eq. (3).

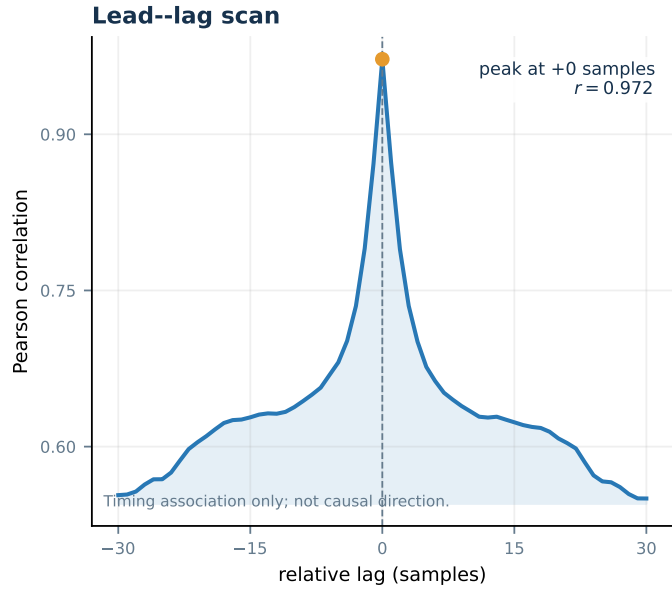


Figure 5: Representative correlation-lag scan for the selected channel pair.

$$\mathbf{a} = g(\mathbf{X}) = \{\mathbf{a}_{\text{health}}, \mathbf{a}_{\text{morph}}, \mathbf{a}_{\text{anom}}, \mathbf{a}_{\text{lag}}, \mathbf{a}_{\text{spatial}}\}, \quad (4)$$

which is passed to retrieval. This separation prevents free-form text from becoming the sole interface between measurement analysis and manual-grounded explanation.

2.2.5. Observation packet, condition report, and structured alert summary

Each completed probe contributes to an observation packet that records the analysis scope, the registered probe and its operational parameters, observed signal events, and relevant uncertainty or data-health cues. The human-readable condition report preserves these results in chronological form for audit and engineering inspection. Its compact alert summary retains only the fields needed downstream—for example, data-health indicators, morphology, spike and zero summaries, co-movement, lag, extremes, and snapshot cues. This report/summary separation preserves auditability without forcing retrieval to operate on a long, unstructured narrative; neither artifact is a fault label or a confirmed diagnosis.

2.3. Report-driven evidence retrieval

The condition report serves two audiences. Its chronological narrative and optional JSON snapshots support human inspection and audit, whereas its compact alert summary is the machine-facing interface to retrieval. The summary contains variable-wise alerts, spike and zero-event counts, morphology, collinearity, lag, and selected spatial-snapshot cues. This descriptive semantic layer lets a reviewer move from a large unlabeled record to a bounded technical question before consulting manuals. Queries are constructed from this structured representation rather than from a short operator phrase or an unreviewed free-text diagnosis hypothesis.

The alert-summary schema constrains retrieval intent without prescribing a fixed set of queries. As shown in table 3, each human-specified cue family is translated into a technical evidence target; wording can then be adapted to the observed record, the query language, and the available terminology. The retrieval intents and query formulations are author-defined and manually reviewed as part of the workflow protocol. This mapping does not claim that queries are generated automatically, and it does not turn an observed cue into a physical mechanism or fault label.

Table 3: Abstract mapping from alert-summary cue families to retrieval intents.

Alert-summary cue family	Retrieval intent	Evidence focus
Spike or isolated-zero event	Signal-integrity and anomalous-measurement context	Definitions, checks, and diagnostic considerations
Co-movement or lag cue	Redundancy, shared context, and timing behavior	Instrumentation and monitoring guidance
Morphology or data-health cue	Stability, noise, availability, and monitoring practice	Data-quality and monitoring guidance
Extremum or cross-channel snapshot	Operating context and cross-channel comparison	Operating guidance and recommended checks

Source PDFs are page-cropped when required and processed with LlamaIndex¹³: they are converted to Markdown, chunked, and embedded in a vector store. At retrieval time, the system first searches for passages relevant to the observed phenomenon or mechanism, then may supplement them with passages on redundancy, topology, monitoring practice, or recommended checks. A memo composer links candidate interpretations to the retrieved passages with numbered citations. It must not convert this support into an automatically confirmed root cause.

This design makes the report the semantic bridge between the measurement and evidence resources introduced in section 2.1. It also explains why the retrieval experiments in section 3 are part of the system contribution: they stress-test whether this evidence layer preserves the intended source policy under controlled corpus expansion, query variation, and chunking.

2.4. Constrained tool use and runtime reliability

The workflow treats runtime reliability as a methodological requirement rather than an implementation afterthought. In a multi-step tool-using workflow, mutable paths, malformed action payloads, unregistered actions, and unconstrained narrative transfer can otherwise break the chain between a measurement record and its evidence-linked output. The final protocol converts these workflow risks into explicit constraints and auditable artifacts, summarized in table 4.

Table 4: Workflow risks, enforced runtime constraints, and auditable artifacts.

Workflow risk	Enforced constraint	Auditable artifact
Path or state drift	Only initialization accepts a raw CSV path; later probes resolve one active record from workspace state.	Active-record reference and probe log
Malformed action payload	Extract one bounded JSON object or array and validate the requested action before execution.	Accepted or rejected action record
Unregistered action	Permit only registered probe identifiers; unknown actions fail closed.	Explicit non-execution event
Narrative overload	Preserve the report for audit but expose only a compact alert summary to retrieval.	Report-summary linkage
Unsupported memo claim	Require passage-linked, citation-bearing memo output.	Evidence-linked memo

2.4.1. State ownership and constrained actions

The local LLM planner selects the next analysis action, but it does not own the authoritative state. The Python router executes registered analyzers and updates the workspace blackboard. Once initialization has completed, probes cannot silently switch to another raw or sorted file because the active record is resolved from that state. Missing initialization causes the workflow to fail closed. This moves data-path ownership from mutable model strings to a runtime contract.

At the action interface, the router tolerates common presentation noise while retaining a narrow execution boundary. ReAct¹⁴, Gorilla¹⁵, and Toolformer¹⁶ motivate tool-using language-model workflows, but the present protocol narrows their execution boundary: it extracts the first brace-balanced JSON object or array even when the model adds Markdown fences or short commentary, then validates that the `action` field names a registered probe. The protocol does not claim perfect structured generation; instead, it makes malformed or invented actions visible and non-executable.

2.4.2. Evidence-constrained output boundary

The runtime constraints continue beyond tool execution. A multi-page condition narrative is useful for audit but unsuitable as the sole retrieval query, because it mixes many observations with incidental detail. The system therefore exposes a compact alert summary to the retrieval layer and requires memo claims to remain linked to retrieved passages. Hallucination surveys¹⁷ motivate this guardrail against fluent but unsupported interpretation. The automated chain produces observations and evidence-linked candidate interpretations; an engineer remains responsible for judging context, support adequacy, and any physical conclusion before use.

3. Experimental Design and Retrieval Evaluation

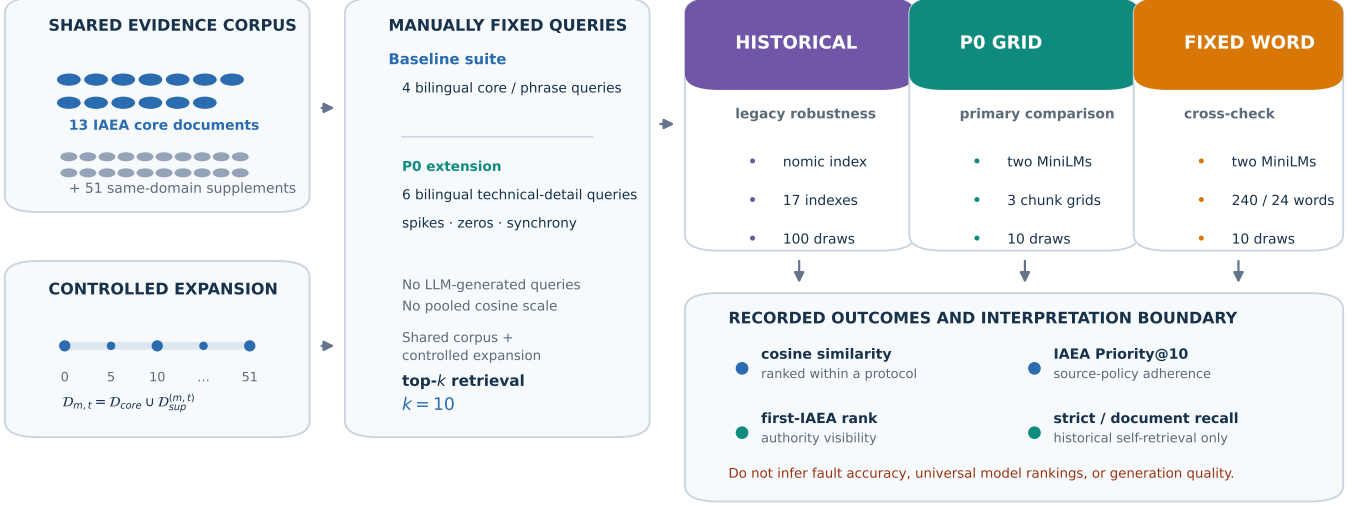


Figure 6: Retrieval evaluation design. A shared 64-document corpus and controlled expansion policy feed manually fixed query suites into three separately interpreted protocol families. The recorded metrics characterize retrieval and source-policy behavior; they do not estimate fault accuracy or generation quality.

Table 5: Retrieval protocol families. Metrics are interpreted within, not pooled across, protocol families.

Protocol	Encoder / index	Chunk construction	Queries and sampling	Primary role
Historical robustness	nomic-embed-text ; 17 saved indexes	Character chunks: 800–1500 characters; valid overlaps 50–150	Four baseline bilingual queries; 100 non-zero draws	Legacy robustness and self-retrieval
P0 chunking grid	Multilingual MiniLM and English-oriented MiniLM	Character chunks: 800/80, 1200/120, 1500/150	Ten bilingual queries; 10 non-zero draws	Query–language–chunking interaction
Fixed-word cross-check	Same two MiniLM encoders	240 words, 24-word overlap	Four baseline queries, plus a small technical-detail pilot; 10 non-zero draws	Independent tokenization check

3.1. Evaluation questions and scope

The evaluation assesses the reliability of the *evidence-retrieval layer*, not diagnostic accuracy, root-cause identification, or generation-side factuality. It asks whether retrieval maintains a deployment policy that prioritizes a fixed IAEA core when same-domain supplementary manuals are admitted, and how that behavior changes with corpus scope, query formulation, chunking, and encoder choice.

The retrieval experiments are deliberately separated from generative-LLM behavior. Query texts are manually specified formulations, and retrieved chunks are ranked through embedding similarity; the experiments do not evaluate LLM-generated queries, tool-use trajectories, memo quality, or generation-side factuality. This separation makes the reported source-priority and self-retrieval metrics attributable to retrieval configuration rather than to variation in a generative model’s wording or reasoning.

Figure 6 provides the chapter-opening map of the shared corpus, manually fixed query suites, protocol families, and recorded outcomes; table 5 records the corresponding configuration boundaries before the detailed definitions.

3.2. Corpus, source policy, and protocol variants

The retrieval collection contains 64 technical documents: a fixed 13-document IAEA core and 51 same-domain supplementary documents. The supplementary material is relevant nuclear instrumentation and control evidence, not meaningless noise. The source-priority policy therefore does not assert that IAEA passages are the only relevant passages; it asks whether a deployment that requires a pre-designated authority set can maintain that preference as the searchable corpus expands.

Three non-pooled protocol families are used. As summarized in the opening overview (fig. 6 and table 5), the historical **nomic-embed-text** index supplies the broadest

legacy robustness and self-retrieval results. The primary query-formulation-by-chunking comparison, hereafter P0, reuses the historical character-based chunking regime at three scopes and evaluates two MiniLM encoders with both baseline and manually authored technical-detail queries. Here, P0 is an experiment-priority label rather than a model or processing stage. The 240-word MiniLM setting is retained last as a fixed-word cross-check, rather than being treated as the reference scale for the other protocols.

Because the encoders and chunk boundaries differ, an absolute cosine value from one protocol is not compared directly with a value from another. Comparisons instead use the source-priority trajectory, rank summaries, and within-protocol differences.

3.3. Queries and controlled corpus expansion

The baseline suite contains four manually fixed bilingual queries: short core-term and measurement-phrase formulations in Chinese and English. P0 additionally tests six manually authored technical-detail queries, three in each language, describing repeated spikes, synchronized zero-value dropouts, and high cross-channel synchrony without time lag. The exact bilingual strings are reported in appendix Appendix A. The query suite and the retrieval intents it operationalizes are author-defined and manually reviewed protocol choices, rather than labels or LLM-generated outputs. Chinese strings are represented by Unicode code points in the appendix solely because the manuscript is compiled with `pdfLaTeX`; the benchmark metadata stores and submits the corresponding UTF-8 strings. The suite intentionally includes both short and specific phrasings, so the study evaluates query wording and language under a fixed corpus rather than claiming an intrinsic language ranking.

The technical-detail wording is derived from cue families that the workflow preserves in a condition report, but it is not automatically generated. These formulations test whether adding a particular technical detail changes the source-priority outcome; they do not assert that longer or more detailed queries are uniformly better.

Let $\mathcal{D}_{\text{core}}$ be the fixed IAEA pool and let $\mathcal{D}_{\text{sup}}^{(m,t)}$ denote the supplementary documents sampled for expansion size m and trial t . As defined in eq. (5), the active retrieval pool is

$$\mathcal{D}_{m,t} = \mathcal{D}_{\text{core}} \cup \mathcal{D}_{\text{sup}}^{(m,t)}, \quad m \in \{0, 5, 10, \dots, 51\}. \quad (5)$$

For $m = 0$, the core-only pool is evaluated once. The historical index experiments use 100 random supplementary-document draws for every non-zero m , query, and index configuration. P0 and the fixed-word MiniLM cross-check each use ten draws. Retrieval returns the top $k = 10$ chunks.

3.4. Historical robustness and self-retrieval experiments

The historical corpus-scope experiment records top-1, top-3 mean, and top-10 mean cosine scores, IAEA prior-

ity@10, and the rank of the first IAEA result. For normalized query and chunk embeddings, cosine similarity is defined in eq. (6), and chunks are ranked in descending order of this score:

$$s(q, d) = \frac{\mathbf{e}(q)^\top \mathbf{e}(d)}{\|\mathbf{e}(q)\|_2 \|\mathbf{e}(d)\|_2}. \quad (6)$$

The primary score plotted in fig. 8 is the top-1 cosine score. For a query q and ordered retrieval list $R_k(q) = \{d_1, \dots, d_k\}$, source priority is defined in eq. (7):

$$\text{Priority}@k(q) = \frac{1}{k} \sum_{i=1}^k \mathbb{I}(d_i \in \mathcal{D}_{\text{core}}). \quad (7)$$

Priority@k measures adherence to a source-priority policy, not human-judged relevance.

Self-retrieval samples IAEA chunks as probes and forms queries using four modes: `first_sentence`, `title`, `random_window`, and `domain_term`. For the corpus-decay condition visualized in fig. 17, the first-sentence mode is used. The core-only pool uses 80 probes. Each non-zero expansion size uses ten trials of 80 probes each. For a probe chunk n_j^* , its source document $d_j^* = \text{doc}(n_j^*)$, and retrieved set $R_k(q_j)$, strict and document-level recall are defined in eq. (8):

$$\begin{aligned} R_k^{\text{strict}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(n_j^* \in R_k(q_j)), \\ R_k^{\text{doc}} &= \frac{1}{N} \sum_{j=1}^N \mathbb{I}(\exists n \in R_k(q_j) : \text{doc}(n) = d_j^*). \end{aligned} \quad (8)$$

We additionally report mean reciprocal rank in eq. (9), where r_j is the rank of the first relevant result and $r_j = \infty$ when the top- k list has no relevant result:

$$\text{MRR} = \frac{1}{N} \sum_{j=1}^N \begin{cases} 1/r_j, & r_j \leq k, \\ 0, & r_j > k. \end{cases} \quad (9)$$

The strict/document distinction prevents overlap-induced neighboring chunks from being characterized as wholly irrelevant.

3.5. P0 query-formulation-by-chunking comparison

P0 is the principal new controlled MiniLM comparison. It holds the 64-document collection, top-10 retrieval rule, source-policy metric, and ten-query suite fixed while varying three historical character-chunk settings (800/80, 1200/120, and 1500/150) and two MiniLM encoders. This design exposes interactions among query family, language, chunking, and encoder without mixing their score scales with the historical `nomic-embed-text` index.

For each model-chunk setting, the protocol records top-1 and top-10 mean cosine scores, IAEA priority@10, and first-IAEA rank for every scope and random draw. The core-only value at $m = 0$ is deterministic under this

construction. At $m = 51$, every supplementary document is present, so a zero across-draw standard deviation reflects an identical full pool rather than evidence of robust retrieval. The primary P0 visualizations therefore compare priority at $m = 10$, its change through $m = 51$, and the complete scope trajectory.

3.6. Fixed-word MiniLM cross-check

The fixed-word comparison is retained as a separate cross-check after P0. It contrasts `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` with `sentence-transformers/all-MiniLM-L6-v2` using 240-word chunks, 24-word overlap, normalized embeddings, and the same 64-document collection. The baseline benchmark uses the four queries in appendix Appendix A; a small companion pilot uses the six manual technical-detail queries documented there on the already saved fixed-word chunks. Both use ten random supplementary-document draws for non-zero scope values.

This protocol is a direct measured check, not an extrapolation from the historical index. It tests whether an observed source-priority risk is confined to one tokenization scheme, but its absolute cosine scale and chunk count are intentionally not merged with P0.

3.7. Aggregation and interpretation boundary

Within each historical index configuration, stored standard deviations quantify variation across random supplementary-document draws. The error bands in figs. 8 and 17 have a different meaning: each configuration is first reduced to its mean, and the displayed bands are one standard deviation across the 17 available chunking configurations. For P0 and the fixed-word MiniLM cross-check, standard deviations are computed over their ten random draws for each query–encoder–chunk–scope condition. These distinct sources of variation are not pooled.

The experiments establish retrieval behavior for the corpus, encoders, query sets, and index grids studied here. They do not establish a universal ranking of languages, technical-detail phrasing, chunking strategies, or encoders; human relevance of every retrieved passage; or the factual correctness of generated memos. Those questions require relevance annotation and generation-side review.

4. Results and discussion

4.1. Traceable semantic handoff: from observation to evidence

At archive scale, the useful intermediate result is not an exhaustive fault label for every signal segment but a bounded, auditable description that tells a reviewer what to investigate next. Across representative workflow executions, the measurement-side probes produce condition reports containing spike episodes, isolated zero events, morphology differences, and strongly co-moving channel groups. These

Table 6: Representative event summary generated from a condition-report output. Counts describe detected signal events, not confirmed equipment faults.

Channel suffix	Spike episodes	Isolated zero events
2011	30	12
2012	34	12
2013	32	12
2014	33	12
2015	33	12
2016	33	12
2017	32	12
Mean	32.4	12.0

are descriptive signal observations that can be inspected, preserved in an alert summary, and mapped to the technical evidence targets in table 3; they are not independently confirmed equipment faults or root causes.

Figure 7 and table 6 illustrate the form of this semantic handoff for one representative condition-report output. Its seven channels contain 30–34 detected spike episodes (mean 32.4) and each contains 12 isolated zero events. The repeated zero timestamps across channels are a concrete reason to retain cross-channel timing in the description and retrieve signal-integrity or instrumentation context before advancing a physical interpretation. The figure is therefore evidence of an auditable observation inventory, not an archive-wide event-prevalence estimate.

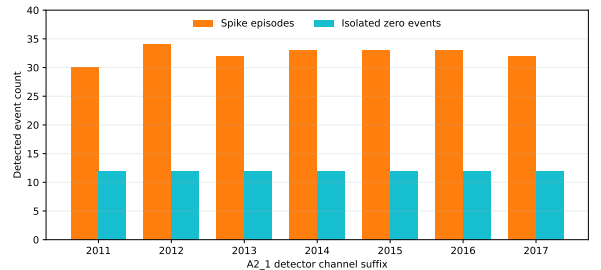


Figure 7: Detected event counts from a representative condition-report output. These are deterministic signal-event counts under eq. (2), not confirmed equipment-fault counts.

4.2. Similarity and source priority can diverge

The corpus-scope results show a consistent deployment risk: increasing the supplementary pool can raise cosine scores while reducing IAEA priority@10. Figure 8 and table 7 aggregate the phrase-query results over all 17 available index configurations. For Chinese, mean IAEA priority@10 falls from 1.00 in the IAEA-only pool to 0.10 ± 0.09 after five supplementary documents and to 0.00 after ten, while the mean top-1 cosine rises from 0.55 to 0.63. English is more gradual: priority@10 remains 0.86 ± 0.05 after five documents and is 0.38 ± 0.05 at 45 documents, while the corresponding cosine stays near 0.69–0.70.

Table 7: Corpus-scope results for phrase queries, averaged across the 17 available index configurations. IAEA priority@10 is a source-priority measure, not a general relevance judgment.

Query language	Supplementary documents	IAEA priority@10	Mean top-1 cosine
English	0	1.00 $\pm$ 0.00	0.69 $\pm$ 0.00
English	5	0.86 $\pm$ 0.05	0.69 $\pm$ 0.00
English	10	0.90 $\pm$ 0.02	0.69 $\pm$ 0.00
English	45	0.38 $\pm$ 0.05	0.70 $\pm$ 0.01
Chinese	0	1.00 $\pm$ 0.00	0.55 $\pm$ 0.00
Chinese	5	0.10 $\pm$ 0.09	0.62 $\pm$ 0.01
Chinese	10	0.00 $\pm$ 0.00	0.63 $\pm$ 0.01
Chinese	45	0.00 $\pm$ 0.00	0.67 $\pm$ 0.00

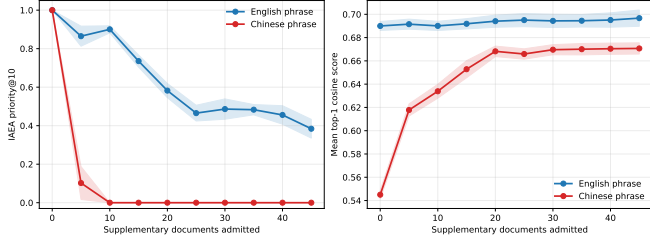


Figure 8: Phrase-query corpus-scope results averaged across 17 index configurations; shaded bands show one standard deviation across configurations. Source priority can decline while the nearest-neighbor cosine score rises.

The aggregate curves conceal substantial configuration-level structure. Figures 9 and 10 retain all 17 individual index configurations and plot the joint change in cosine score and IAEA priority relative to the core-only pool. Together they confirm the same deployment-risk pattern without averaging away the configuration-level detail.

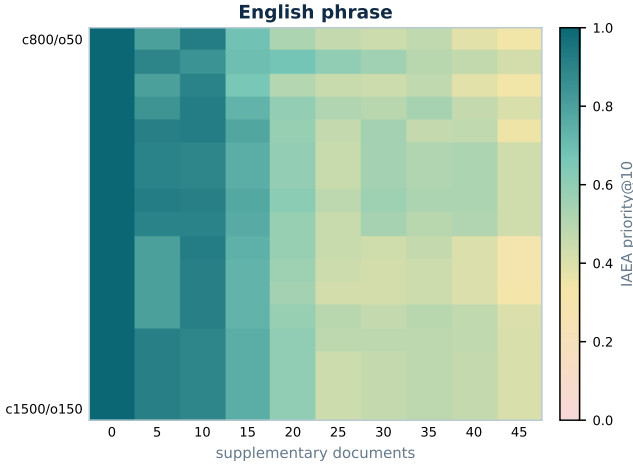


Figure 9: English IAEA priority@10 across individual historical index configurations and supplementary-corpus sizes.

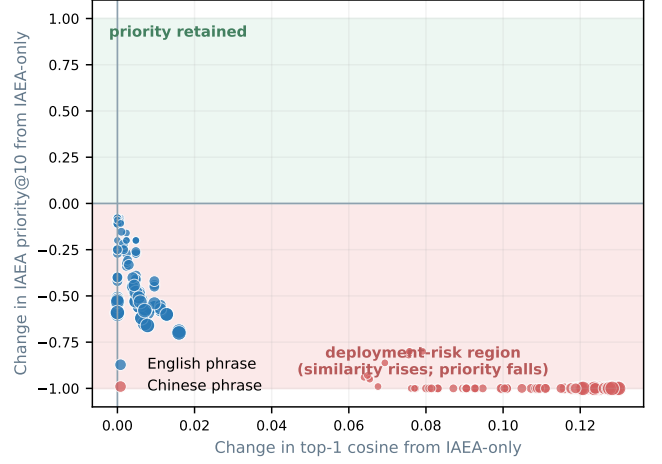


Figure 10: Bilingual retrieval-reliability quadrant. Each point is a configuration–corpus condition positioned by its change from the core-only pool; the lower-right quadrant marks conditions in which similarity rises while source priority falls.

These results do not imply that supplementary documents are irrelevant. Rather, when an application requires IAEA sources to be surfaced first, nearest-neighbor similarity alone is not enough to enforce that policy. Source metadata, filtering, or reranking should be considered before a generated memo is presented as authority-backed evidence.

4.3. Query formulation interacts with language and chunking

The P0 results make the interaction visible rather than reducing it to a single encoder or language ranking. Figures 11 and 12 first isolate the effect of replacing the baseline family with manually authored technical-detail queries at $m = 10$. Each point is the difference between the family means under one encoder–chunk setting. The English differences range from -0.03 to $+0.14$, whereas Chinese differences range from -0.23 to $+0.02$. Thus technical detail improves source priority in some settings and degrades it in others; it is not a generally beneficial query expansion.

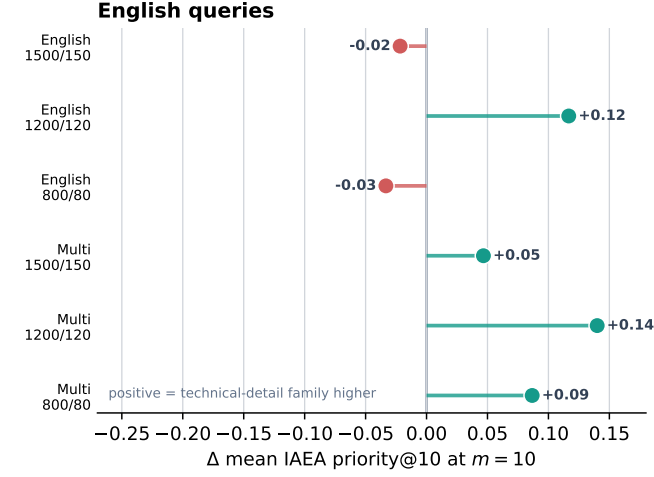


Figure 11: English P0 query-formulation effect at ten admitted supplementary documents. Each point is the technical-detail family mean IAEA Priority@10 minus the baseline-family mean for one encoder-chunk setting.

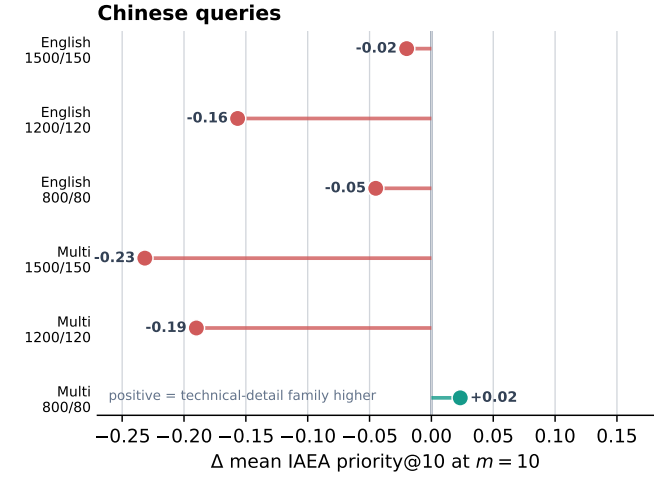


Figure 12: Chinese P0 query-formulation effect under the same settings. Positive values favor the technical-detail family; the sign changes across language and configuration.

The complete ten-query landscape in figs. 13 and 14 provides the configuration-level values behind this summary. English measurement phrases and some technical-detail queries retain high IAEA priority in several conditions, while the same intervention is not uniformly beneficial in Chinese. When averaged over the six model-chunk settings at $m = 10$, the technical-detail English queries have mean priority@10 of 0.65, compared with 0.60 for the English baseline queries; the corresponding Chinese values are 0.52 and 0.62. These are descriptive aggregates over the recorded settings, not evidence that a particular language or longer formulation is universally preferable.

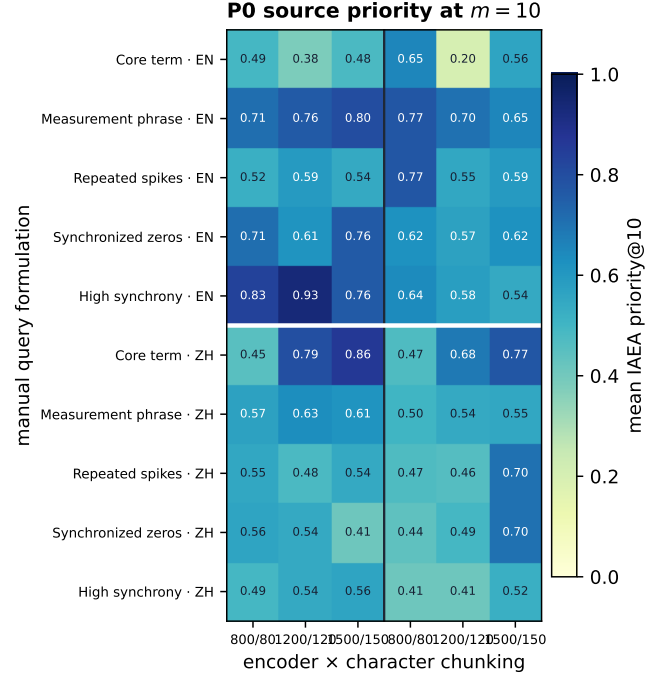


Figure 13: Mean P0 IAEA priority@10 at ten admitted supplementary documents for every manually fixed query and model-chunk setting. Technical-detail queries are manually authored from preserved observation cues, not LLM-generated.

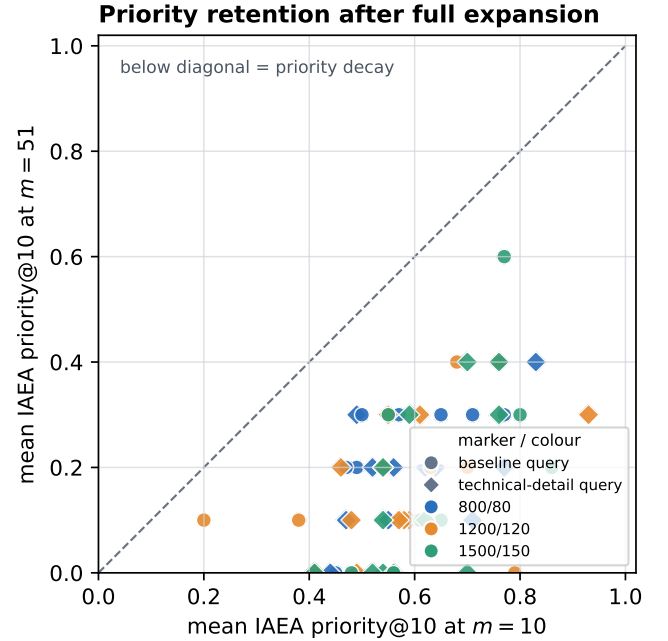


Figure 14: P0 priority retention after all 51 supplementary documents are admitted. Points below the diagonal lose source priority relative to $m = 10$.

Figures 15 and 16 add the full corpus-expansion paths for the English and Chinese queries. Every curve begins at one in the IAEA-only pool by definition, then generally falls as supplementary documents enter the retrieval pool. The

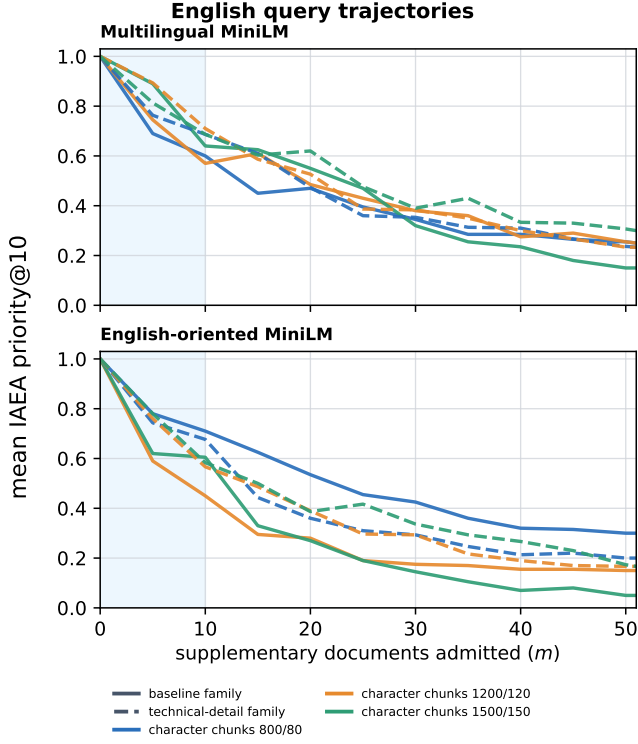


Figure 15: English P0 source-priority trajectories over corpus expansion for both encoders. Colour denotes character chunking and line style denotes the manually fixed query family.

separation and occasional crossings between solid baseline-family and dashed technical-detail-family curves show why the query effect cannot be stated as a universal improvement. The figures also show that the source-priority risk persists across all three historical chunking scopes, even though its magnitude depends on the joint configuration.

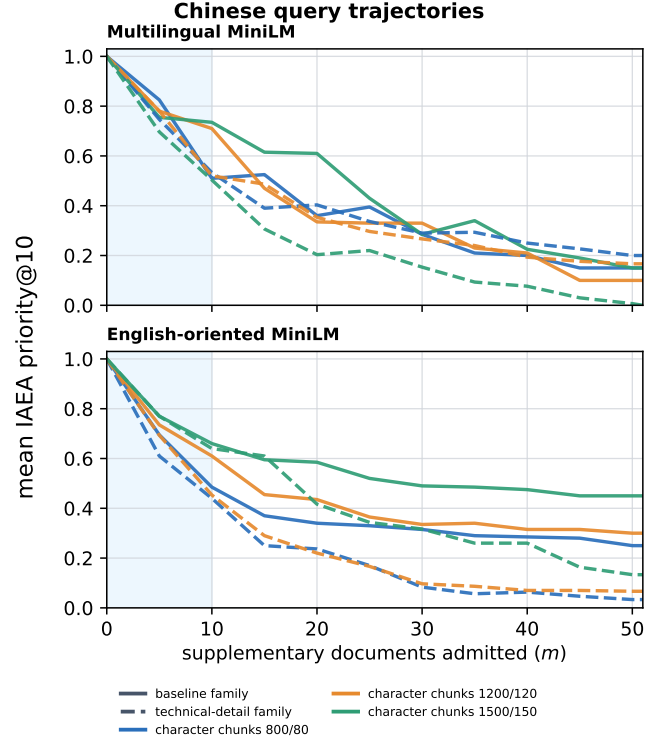


Figure 16: Chinese P0 source-priority trajectories under the same protocol. Curves average queries within a family and use the Priority@10 definition in eq. (7).

4.4. Strict and document-level relevance answer different questions

Self-retrieval exposes a systematic strict-document-level gap. Figures 17 and 18 show the complete corpus-decay curves separately for Recall@10 and MRR so that both panels remain readable within a single column. In one 800/50 setting, IAEA-only Recall@10 is 0.7625 under exact-node matching and 0.8625 under document-level matching. The gap is expected when overlapping chunks preserve relevant content in a neighboring node. Conversely, terminology-rich domain sentences have substantially stronger document-level Recall@10 in the available results. Evaluation should therefore report both criteria: strict matching measures chunk re-identification, while document-level matching better reflects whether the correct manual was recovered.

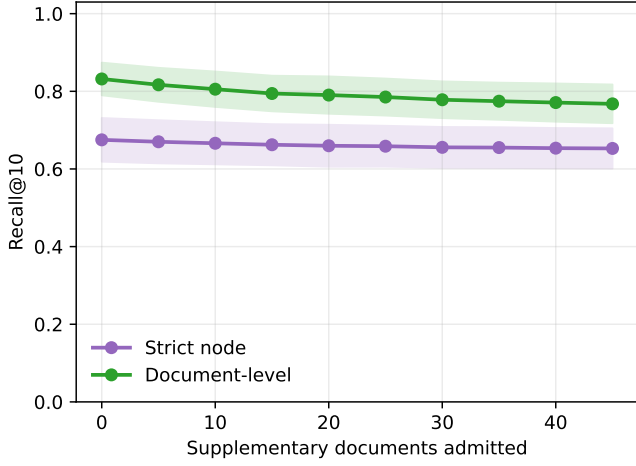


Figure 17: First-sentence self-retrieval Recall@10 under corpus expansion, aggregated over available result directories. Shaded bands show one standard deviation across configurations.

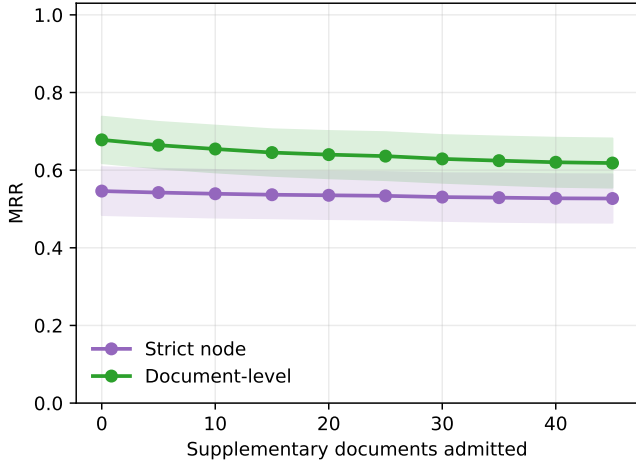


Figure 18: First-sentence self-retrieval MRR under the same corpus-expansion conditions. Document-level relevance remains higher than exact-node matching because overlapping chunks can preserve relevant text within the same document.

4.5. Fixed-word MiniLM cross-check

To test whether the observed behavior is tied only to the historical character-based chunking, we also ran the independent fixed-word MiniLM protocol specified in section 3. It compares a multilingual MiniLM encoder with an English-oriented MiniLM encoder and is reported separately from P0 because the 240-word construction changes the retrieval units.

The language-specific panels in figs. 19 and 20 and the checkpoint values in table 8 show that encoder choice changes cosine scales and source-priority trajectories, but does not collapse the problem into one scalar. The multilingual model gives materially higher Chinese phrase cosine scores (0.62–0.64 versus 0.27 for the English-oriented model), while both models undergo source-priority decay as the supplementary pool grows. Thus multilingual se-

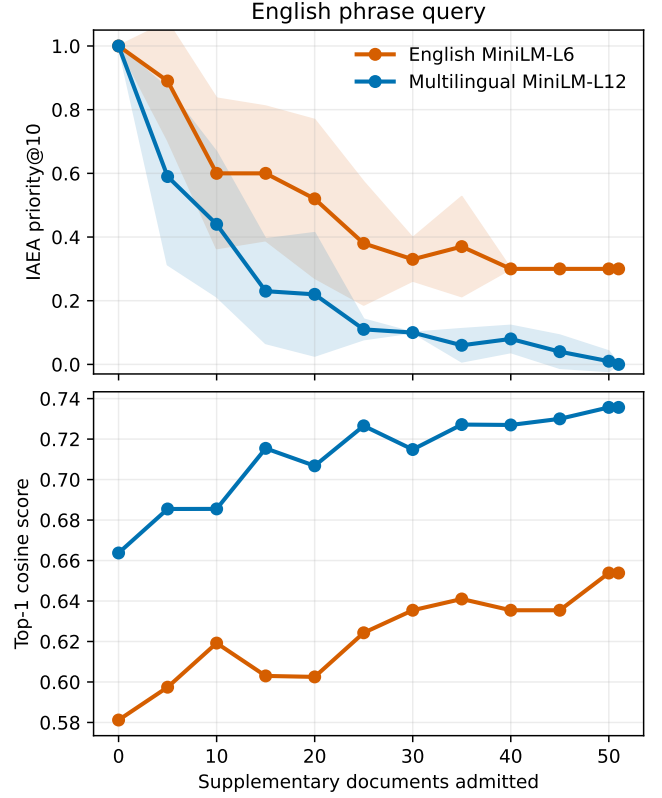


Figure 19: Fixed-word MiniLM cross-check for the English phrase query. The upper panel reports IAEA priority@10 and the lower panel reports top-1 cosine score over corpus expansion.

mantic alignment and authoritative-source preference are complementary deployment requirements rather than interchangeable objectives.

Table 8: Independent embedding-model comparison (64 documents, 13 IAEA core documents, 11,199 chunks, 10 random trials after corpus expansion). This comparison uses 240-word chunks and is reported separately from the historical nomic-embed-text index.

Model	Query language	Supplementary documents	IAEA priority@10	Top-1 cosine
Multilingual MiniLM-L12	English	0	1.00 ± 0.00	0.66
Multilingual MiniLM-L12	English	10	0.44 ± 0.23	0.69
Multilingual MiniLM-L12	English	51	0.00 ± 0.00	0.74
Multilingual MiniLM-L12	Chinese	0	1.00 ± 0.00	0.62
Multilingual MiniLM-L12	Chinese	10	0.53 ± 0.15	0.64
Multilingual MiniLM-L12	Chinese	51	0.10 ± 0.00	0.64
English MiniLM-L6	English	0	1.00 ± 0.00	0.58
English MiniLM-L6	English	10	0.60 ± 0.24	0.62
English MiniLM-L6	English	51	0.30 ± 0.00	0.65
English MiniLM-L6	Chinese	0	1.00 ± 0.00	0.27
English MiniLM-L6	Chinese	10	0.48 ± 0.27	0.27
English MiniLM-L6	Chinese	51	0.20 ± 0.00	0.27

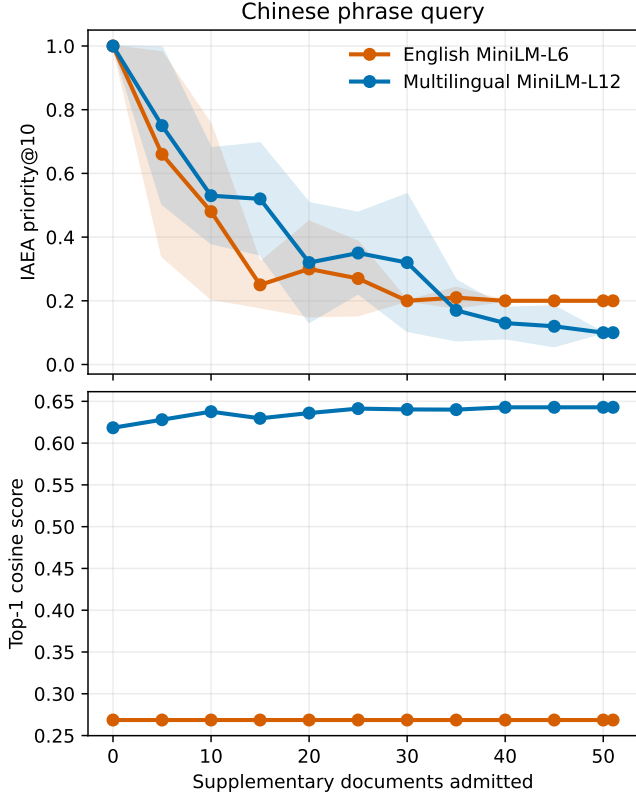


Figure 20: Fixed-word MiniLM cross-check for the Chinese phrase query under the same 64-document protocol. The multilingual encoder raises the cosine scale, but both encoders still exhibit source-priority decay.

4.6. Discussion and deployment implications

The pipeline provides traceable observations and evidence retrieval, but it does not replace engineering judgment. In a deployment where particular source families must be surfaced first, the results support terminology-rich condition-report queries together with source metadata, filtering, or reranking. The remaining validation boundaries and threats to interpretation are stated separately in section 5.

5. Limitations and Threats to Validity

This study evaluates an evidence-constrained assistance workflow, not an autonomous fault classifier. Its principal limits concern the interpretation of the signal observations, the coverage of retrieval evaluation, and the scope of the embedding comparison.

5.1. Absence of ground truth and observation-to-fault inference

The raw measurement CSV records used in this study do not provide a uniformly aligned, gold-standard fault taxonomy. The deterministic probes therefore report signal events—for example, spikes, zero values, extreme values, and cross-channel co-movement—rather than confirmed

equipment faults. Generated condition reports are derived workflow artifacts, not ground-truth annotations. They demonstrate traceability from data to an observation record, but cannot establish end-to-end diagnostic accuracy or root-cause accuracy. Engineering review and, where available, maintenance records remain necessary to determine the physical mechanism behind an observed event.

5.2. Retrieval corpus and relevance coverage

The robustness study is bounded by a fixed 64-document corpus, consisting of a 13-document IAEA core and 51 same-domain supplementary documents. Its corpus-expansion protocol is useful for testing whether designated authoritative sources remain visible, but it is not a representative sample of every plant documentation environment. The query suite and retrieval intents are author-defined and manually reviewed as part of the stated workflow protocol; this is a traceable design choice, not an independent expert relevance assessment. The scope experiments do not include a human relevance judgment for each retrieved passage or an expert audit of every citation-bearing memo claim. Consequently, strict chunk and document-level self-retrieval are retrieval diagnostics, not direct measures of clinical or operational usefulness.

5.3. Embedding and generation boundaries

The historical index results and the independent embedding-model comparison use different chunk constructions and are therefore reported separately. The latter compares two MiniLM encoders under one fixed retrieval configuration; it establishes that encoder choice affects bilingual similarity scales and source-priority decay, but it does not rank all available embedding models. The study also evaluates retrieval rather than generation-side factuality, completeness, or calibration; it does not estimate the effect of changing a generative LLM on query formulation, tool use, or memo quality. Any generated diagnostic memo should remain linked to retrieved passages and be reviewed by a qualified engineer before it informs a safety-relevant decision.

5.4. Threats to validity and next validation stage

The reported relationships between corpus expansion, cosine similarity, and source priority may change with the corpus, metadata policy, chunking implementation, query formulation, encoder, or reranker. Repeated measurements reduce the sensitivity of the reported curves to random supplementary-document selections, but they do not eliminate these external-validity limits. A next validation stage should separately assess observation correctness against reviewed cases, passage-level support for each memo claim, expert usefulness of the memo, and generation-side factuality under a preregistered evaluation protocol.

6. Conclusions

We presented an evidence-constrained workflow for diagnostic assistance on unlabeled neutron-current time series. The workflow uses deterministic analysis probes under a persistent workspace contract to produce auditable observations, then turns a structured alert summary into manual-grounded, citation-bearing explanatory support. Its output is deliberately framed as an aid to engineering review rather than an autonomous fault decision.

The retrieval study identifies a practical governance issue: for the evaluated corpus and embedding model, cosine similarity can improve while the priority of pre-designated IAEA sources declines as same-domain supplementary manuals are added. Short Chinese queries are particularly sensitive in the tested setting, and document-level relevance is more forgiving than exact chunk identity under overlapping segmentation. The separate embedding-model comparison further shows that multilingual semantic alignment and authoritative-source preference are complementary requirements rather than interchangeable objectives.

Together, these results favor terminology-rich report-derived queries and explicit source policies over similarity-only deployment choices. The study’s boundaries, including the absence of a sample-level fault ground truth and limited retrieval and embedding coverage, are stated in section 5. Future work will extend the evaluation with expert-reviewed cases, relevance annotations, broader multilingual retrieval configurations, and generation-side factuality assessment.

Appendix A. Full Query Formulations

This appendix records the exact manually specified query suite used in the corpus-scope experiments. The UTF-8 benchmark files store the Chinese formulations directly; Unicode code points are shown here because the manuscript is compiled with `pdfLaTeX`. The strings are author-defined protocol inputs, not fault labels and not LLM-generated queries.

Table A.1: Exact baseline and manual technical-detail query formulations.

Suite	Cue / form	Chinese UTF-8 representation (Unicode code points)	English formulation
Baseline	Core term	U+4E2D U+5B50	neutron
Baseline	Measurement phrase	U+4E2D U+5B50 U+7535 U+6D41 U+6D4B U+91CF	neutron current measurement
Technical detail	Repeated spikes	U+4E2D U+5B50 U+7535 U+6D41 U+6D4B U+91CF U+91CD U+590D U+5C16 U+5CF0 U+4E8B U+4EF6	neutron current measurement repeated spike events
Technical detail	Synchronized zeros	U+4E2D U+5B50 U+7535 U+6D41 U+6D4B U+91CF U+8DE8 U+901A U+9053 U+540C U+6B65 U+96F6 U+503C U+95EA U+65AD	neutron current measurement synchronized zero-value dropout across channels
Technical detail	High synchrony	U+4E2D U+5B50 U+7535 U+6D41 U+6D4B U+91CF U+9AD8 U+540C U+6B65 U+901A U+9053 U+65E0 U+65F6 U+95F4 U+6EDE U+540E	neutron current measurement highly synchronized channels without time lag

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